Homework 5 Theory Questions

1. b) deallocates the dynamic variable that is pointed to by p.

2.
   a. int* num = new int[10];
   b. for (int i = 0; i < 10; i++) {
      std::cin >> new[i];
      }
   c. delete[] num;

3. *numPtr accesses the value stored at the memory address the pointer is holding, since * is a dereference operator, and &numPtr accesses the memory address the pointer is holding.

4. The given C++ code will not work because nextPtr is set equal to firstPtr, and then firstPtr is deleted, and nextPtr is deleted after. nextPtr cannot be deleted because the memory in firstPtr has already been deleted.

5.
   a. arr = new Number[25];
   b. for (int i = 0; i < 25; i += 5) {
      arr[i].getSign();
      }
   c. delete[] arr;

6. A shallow copy duplicates only the memory address so both pointers share the same data, while a deep copy creates brand new memory and duplicates the actual data so they are completely independent.